Sequence Integrity Matters: Row-Level Client Partitioning Artifacts in Federated RAN Time-Series Benchmarks

Hsiu-Chi Tsai
National Yang Ming Chiao Tung University
nctuie@gmail.com

***Abstract*—Federated-learning (FL) benchmarks routinely synthesise client heterogeneity by partitioning a pooled dataset across clients with a Dirichlet distribution over *rows*. For tabular or image data this is harmless. For *time series*—where each client subsequently builds sliding windows—it is not: row-level partitioning scatters a sequence's consecutive timesteps across clients, so the per-client windows are no longer real trajectories. We show that this *sequence-integrity* violation can manufacture a striking but spurious finding—that *more* client heterogeneity *improves* accuracy—in a federated O-RAN slice-SLA benchmark (ColO-RAN), and we dismantle it. The fragmentation itself is universal (a deterministic consequence of the partition-then-window recipe, confirmed on two real RAN datasets), but its impact on accuracy is task-conditional: it degrades AUC substantially only when the prediction target is *sequence-essential* (a small order-invariant residual remains for window-aggregate targets). We introduce a cheap, partition-free diagnostic, Δ_{traj}—the AUC an intact sequence model gains from temporal *order* over an order-shuffled baseline—and show the fragmentation AUC gap is a monotone function of it (Spearman 0.95 across 11 ColO-RAN targets; Spearman 1.00 / Pearson 0.99 in a controlled synthetic study). The artifact is architecture-invariant for order-using models (LSTM, GRU) and absent for order-blind models, confirming it is sequence-specific. We give a corrected partitioning protocol (partition by entity/run; synthesise heterogeneity with *run-level* Dirichlet) and recommend Δ_{traj} as a pre-submission screen. On a second real dataset (Open RAN Commercial Traffic Twinning) the mechanism replicates but the AUC impact does not—as the diagnostic predicts for its persistent targets—serving as a negative control.**

I. Introduction

Client heterogeneity (non-IID data) is the defining difficulty of FL, and benchmarks emulate it by splitting a pooled dataset with a Dirichlet(α) distribution: small $\alpha \to$ skewed, large $\alpha \to$ near-IID. This recipe is near-universal [1], [2], [3] and, for i.i.d.-sample data (images, tabular rows), statistically sound.

Time-series FL inherits the recipe but adds a step: each client turns its rows into fixed-length sliding windows for a sequence model. Here the recipe quietly breaks. If rows are assigned to clients independently, a run's consecutive timesteps are *scattered* across clients; when a client then windows its rows (sorted by time), each "window" is a set of non-consecutive snapshots, and the label attached to it belongs to a temporally mismatched step. The sequence model is trained on temporally incoherent (window, label) pairs.

We encountered this concretely. In a federated O-RAN slice-SLA prediction benchmark on the ColO-RAN/Colosseum testbed [4], partitioning by Dirichlet over rows produced an *inverted* heterogeneity curve: AUC rose as α fell (more skew $\to$ better accuracy), the opposite of the FL norm—recent large-scale assessments find heterogeneity *harms* accuracy [5]. The tempting reading—that RAN telemetry has special cell-conditional structure—does not survive controls. A **run-level** Dirichlet (whole runs assigned with the same skew) recovers the natural-partition accuracy and erases the inversion; breaking entity coherence while keeping runs intact costs $\sim$0 AUC. The apparent inversion is a **sequence-integrity artifact of how the non-IID partition is constructed**, not a property of the data.

Contributions. (1) We identify and characterise the *sequence-integrity pitfall* of row-level partitioning in time-series FL benchmarks, with a simple *fragmentation audit metric*. (2) We show the AUC impact is **task-conditional** and introduce $\Delta_{\mathrm{seq}}/\Delta_{\mathrm{traj}}$, capacity-matched, partition-free diagnostics that *predict* the gap. (3) We give the mechanism (run-rate vs. trajectory; order vs. consecutiveness) and a careful theory: the diagnostic is monotone, *not* a bound (the natural gap $\leq \Delta_{\mathrm{seq}}$ conjecture is falsified). (4) We establish architecture-invariance (LSTM, GRU) and order-blind sanity controls. (5) We give a corrected protocol and a negative control on a second real RAN dataset.

II. Related Work

(a) The Dirichlet partitioning convention. Simulating non-IID clients by a Dirichlet distribution over labels/rows is the de-facto FL-benchmark standard—codified by NIID-Bench [2] and shipped as the default partitioner in FL frameworks [3]. It is sound for *i.i.d.-sample* data; our point is that time-series FL *inherits* it and then adds per-client windowing, where it silently breaks. The "more heterogeneity helps" reading we dismantle is especially suspect because the established norm is the opposite—heterogeneity *harms* [5]—so an *inverted* curve should itself be a red flag for an artifact.

(b) Windowing leakage in time series. A known pitfall is generating windows *before* the train/test split, leaking future context into training [6]. This is *a different failure mode*: centralized train/test causality (it inflates scores), not federated client partitioning. Ours is within-training, federated-specific, and *degrades*—about per-client window integrity across clients.

TABLE I
FRAGMENTATION AUDIT (FRACTION OF CONTIGUOUS WINDOWS).

dataset	intact	random_split	dirichlet
ColO-RAN	1.000	$\sim$0	$\sim$0
Twinning	1.0000	0.0002	0.0031

(b′) Temporal-order ablation (our Δ_{traj} method has precedent). Shuffling timesteps to probe a model's reliance on temporal order is established—temporal-order verification [7] and segment-shuffle representation learning [8], which notes non-adjacent timesteps can carry strong dependencies. We do *not* claim the shuffle ablation as novel; we reuse it for a new purpose—predicting/screening the federated fragmentation gap—and show (Sec. V) that the shuffle destroys *order* whereas row-level fragmentation destroys *consecutiveness*.

(c) Temporal fragmentation as a deployment challenge. Some FL works note local time-series can be "fragmented" and that this degrades feature extraction. We differ: the fragmentation here is *manufactured by the row-level recipe as a benchmark artifact*, is *task-conditional* with a *predictive diagnostic*, and can produce a *qualitatively false finding*. FedSL [9] instead studies legitimately sequentially-split sequences—a different setting.

(d) FL time-series / RAN benchmarks. FL traffic/throughput forecasting [10], [11], [12], O-RAN toolchains [13], [4], [14], and FL forecasting benchmarks [15] are the landscape. Susceptibility hinges on the partition-then-window order (Sec. VI): entity-partitioned benchmarks are safe; row-Dirichlet ones are at risk.

III. The Sequence-Integrity Pitfall

A pooled series is split into clients, then each client builds length-L sliding windows within its rows (sorted by step). *Intact* partitions (natural entity, or run-level Dirichlet) keep whole groups together $\rightarrow$ genuine trajectories. *Row-level* partitions scatter a group's rows $\rightarrow$ fragmented windows.

Audit metric. The fragmentation score = fraction of per-client windows with contiguous `step_idx`: 1.0 for intact, $\sim$0 for row-level on both datasets we study.

The audit is *necessary but not sufficient* for an AUC impact—a window can be fragmented yet the task unharmed (Sec. IV). Fragmentation alters within-window *order*, *which* steps populate a window, and the window-to-label *alignment*.

IV. The Δ_{seq} / Δ_{traj} Diagnostic and Mechanism

Diagnostics (capacity-matched, partition-free). Δ_{seq} = AUC(seq LSTM) $-$ AUC(same LSTM, length-1 window): value of the multi-step window over a single step. Δ_{traj} = AUC(seq LSTM) $-$ AUC(same LSTM, order-shuffled windows): isolates the *order/trajectory* value (the partition-vulnerable part); the cleaner predictor.

Gap decomposition. The gap splits into (i) a *trajectory component* (only sequence models; $\propto \Delta_{\mathrm{traj}}$; dominant for white-noise targets) and (ii) a small *window-content component* (any model, only for *aggregate* targets). A no-sequence mean-pool MLP isolates (ii): 0.000 for a point target, a real 0.025 for a 5-step-smoothed target ($\ll$ the LSTM's 0.082).

TABLE II
SOURCE-COLUMN LAG-1 AUTOCORRELATION (COLO-RAN).

source	autocorr	nature	Δ_{traj}/gap
tx_brate_dl	0.984	run-rate	$\sim$0
dl_buffer	0.999	run-rate	$\sim$0
dl_mcs	0.902	run-rate	$\sim$0
dl_cqi	0.553	moderate	$\sim$0
ul_bler	**0.022**	**trajectory**	**0.22/0.15**

TABLE III
COLO-RAN Δ_{seq} LAW (11 TARGETS).

target	Δ_{seq}	Δ_{traj}	gap
buffer_med	0.013	0.000	+0.000
mcs_med	0.039	0.002	+0.000
cqi_med	0.048	0.001	−0.011
brate_med	0.111	0.010	−0.008
bler_trend5	0.159	0.060	+0.083
bler_k2	0.170	0.200	+0.134
bler_k3	0.209	0.219	+0.121
bler_th20	0.221	0.222	+0.149
bler_th10	0.227	0.225	+0.152
bler_th05	0.227	0.225	+0.152
bler_k5	0.233	0.232	+0.158

Mechanism: run-rate vs. trajectory. A target's predictability splits into a partition-invariant *run-rate* (estimable order-free from any sample) and a partition-vulnerable *trajectory*. The source's lag-1 autocorrelation is a pre-diagnostic (Table II).

A diagnostic, not a bound. The gap is monotone in Δ_{traj} but we do *not* claim a bound. The conjecture gap $\leq \Delta_{\mathrm{seq}}$ is *falsified*: with a pure-trajectory synthetic target, fragmented training is mis-led below the single-step ceiling (gap $> \Delta_{\mathrm{seq}}$), whereas real BLER (with a run-rate fallback) gives gap $< \Delta_{\mathrm{seq}}$.

V. Experiments

5.1 Synthetic causal control. Multivariate AR runs; a knob λ interpolates a target between instantaneous and trajectory dependence; pos-rate held $\sim$0.5, all else fixed. The gap is monotone in Δ_{seq} (Pearson 0.986) and the gap $\leq \Delta_{\mathrm{seq}}$ bound is falsified at high λ.

5.2 ColO-RAN Δ_{seq} law (main result). 11 targets, 5 seeds, OOD-by-tr split (Table III). The gap is monotone in Δ_{seq} (Pearson 0.943, Spearman 0.918, OLS slope CI95 $[0.706, 1.113]$ excludes 0) and cleaner in Δ_{traj} (Pearson 0.974, Spearman 0.945). BLER targets: gap 0.08–0.16 (5-seed paired-bootstrap CIs exclude 0); CQI/MCS/buffer/throughput: gap $\sim$0. `bler_th10` reproduces the established ColO-RAN gap (intact 0.908 / row 0.757). The law and its Δ_{traj} refinement are shown in Fig. 1.

5.3 Architecture invariance + sanity. A no-sequence mean-pool MLP shows gap $\sim$0 even for high-Δ_{seq} BLER

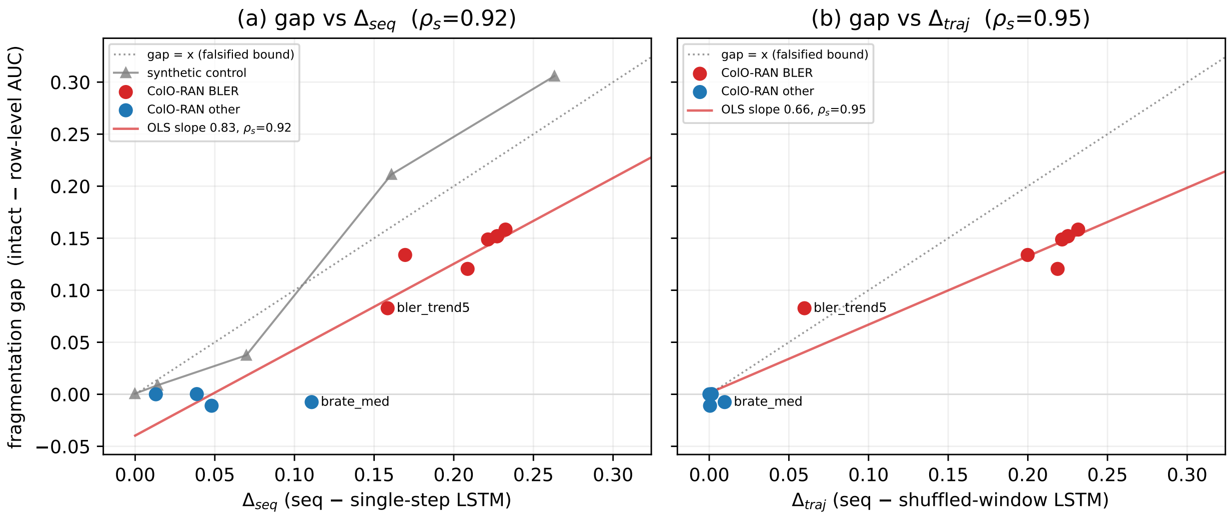

Fig. 1. The fragmentation AUC gap (intact − row-level) is a monotone function of the sequence-essentiality diagnostic. **(a)** vs. Δ_{seq} (seq − single-step LSTM), with the synthetic controlled backbone and the gap $= x$ line (the *falsified* bound). **(b)** vs. the order-isolating Δ_{traj} (seq − shuffled-window LSTM): tighter (ρ_s=0.95) and pulls the persistent `brate_med` deviator onto the origin. Order-free targets (CQI/MCS/buffer/throughput) cluster at gap≈0; BLER targets rise with the diagnostic. Slopes < 1 in (b) reflect $\Delta_{\text{traj}} \geq$ gap (it over-predicts; Sec. V).

(max 0.025, an order below the LSTM's 0.16) $\rightarrow$ the gap is sequence-specific. LSTM and GRU both learn the BLER trajectory (intact ≈0.90) and show the gap tracking Δ_{seq}. Our small Transformer has no positional encoding, so self-attention with mean-pooling is permutation-invariant—it is *order-blind by construction* (intact BLER 0.69 ≈ MLP 0.66) and acts as a second no-sequence sanity: both order-blind models show a negligible BLER gap (MLP +0.0001, Transformer +0.006), an order of magnitude below the order-using LSTM/GRU (+0.16/+0.15). A positional-encoded Transformer is untested (attention's permutation-invariance impeding order modeling is a known limitation).

5.4 Twinning negative control + hunt. On Open RAN Commercial Traffic Twinning [14] (entity = UE; real Madrid LTE traffic twinned via Colosseum), the mechanism replicates (audit intact 1.0 vs. row ∼0, 6858 (run,UE) groups, 31M rows) but the AUC impact does not (next-step CQI/MCS gap ∼0). A hunt for a *within-Twinning* sequence-essential positive (drop-event targets) found targets that are Δ_{traj}-positive (0.07–0.13) yet show no gap (+0.002): row-level windowing keeps each window sorted-ascending (only gappy), so it **preserves order and destroys only consecutiveness** (the row AUC tracks the ordered seqC, not the shuffled baseline), and a drop-event needs order but tolerates gaps. So $\Delta_{\text{traj}} \geq$ gap empirically; it conflates order with consecutiveness. Even Twinning's sequence-essential targets are fragmentation-robust.

VI. A Correct Partitioning Protocol

(1) Partition by natural *entity/run* (each holds its intact stream). (2) To synthesise heterogeneity, use **run-level Dirichlet** (whole runs, skewed assignment), never row-level. (3) Pre-screen with Δ_{traj}, then *confirm with the run-level-vs-row-level control*: a $\Delta_{\text{traj}} \approx 0$ benchmark is safe; a large Δ_{traj} flags a benchmark to investigate, but Δ_{traj} can over-predict (it destroys the last-step/order signal that END-aligned fragmentation preserves), so the partition control is the decisive test.

TABLE IV
Spot-audit by stated partition recipe.

benchmark	stated partition	susceptibility
Beijing air-quality FL [15]	entity (per station), windows post-partition	safe
Mobile-net traffic FL [10]	entity (per BS)	safe
5G BS traffic [11]	entity (per BS)	safe
NIID-Bench / ProFed [2], [16]	Dirichlet over rows (image/tabular)	convention's home; unsafe if transplanted to a pooled TS
ColO-RAN `fl_v7` (this work)	Dirichlet over rows of a pooled series	at-risk (manufactured the inverted-α artifact)

A spot-audit of representative FL-TS benchmarks (Table IV; we classify the stated recipe—a definitive verdict needs each paper's code) shows well-designed *domain* benchmarks predominantly partition by *entity* (safe); the risk is transplanting the generic image/tabular Dirichlet-over-rows convention onto a pooled time series. The partition-then-window *order is rarely reported*, so susceptibility often cannot be determined from the text—itself the finding motivating our reporting recommendation. We do not assert any prior benchmark is flawed.

VII. Limitations and Threats to Validity

The $\Delta_{\mathrm{seq}}/\Delta_{\mathrm{traj}}$ law is an *empirical* monotone diagnostic, not a proven bound (synthetic super-linearity vs. real sub-linearity shows the constant is data-dependent). Δ_{traj} *over-predicts* for order-but-gap-robust targets (Sec. V); it is a cheap screen, and the run-level-vs-row-level control is ground truth. Architecture coverage is LSTM + GRU (order-using) plus a mean-pool MLP and a no-positional-encoding Transformer (both order-blind sanities); a positional-encoded Transformer is untested. The Twinning AUC-impact null is a 1-seed smoke; ColO-RAN gaps use 5-seed paired bootstrap (n=5). `ul_sinr` was dropped pre-results (degenerate: median at a mass point → undefined AUC). Partition client counts differ by mode (iid = 7 BS; Dirichlet = 8), immaterial to the global-test gap.

VIII. Conclusion

Row-level Dirichlet partitioning—a near-universal FL-benchmark convention—can manufacture spurious findings in time-series FL by fragmenting per-client windows. The pitfall is real but *task-conditional*: it bites sequence-essential tasks (large Δ_{traj}) and spares order-free ones, which is why it has gone unnoticed. We provide the audit metric, the Δ_{traj} diagnostic, the mechanism, and a corrected protocol. We urge time-series FL benchmarks to partition by entity/run (or run-level Dirichlet) and to report Δ_{traj} and the partition-then-window order.

Reproducibility

Pre-registration, results, theory, and scripts (synthetic; ColO-RAN sweep; Δ_{traj} addendum; architecture invariance; Twinning audit + AUC-impact + hunt; `window_cache` with a bit-exact test) are in the project repository. All runs: local RTX 4060 Ti (memory-caged) + a V100 ColO-RAN sweep.

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